

Beatrix Hollowell

AI Engineer | E-Commerce Intelligence

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EXPERIENCE

AI Engineer, Vertex Commerce

Seattle, WA | March 2024 – Present

- Developed multi-agent retrieval-augmented generation (RAG) framework for product recommendation synthesis, integrating LLaMA 2 70B with semantic search over 2M product embeddings
- Fine-tuned open-source LLM on curated e-commerce domain corpus (47K query-response pairs); achieved 89.2% accuracy on internal eval set, 12-point improvement over base model
- Implemented agentic workflow orchestration using LangGraph to handle product comparison, inventory lookup, and pricing logic in sequence; measured latency at 340ms per request on benchmarking hardware
- Engineered vector store integration with Weaviate for real-time embedding search; optimized retrieval speed to sub-200ms for top-10 product ranking

Machine Learning Engineer, Merch Dynamics

Portland, OR | July 2023 – February 2024

- Built evaluation framework for comparing LLM outputs on retail-specific tasks; developed 15K-example benchmark dataset spanning product descriptions, customer Q&A, and inventory predictions
- Implemented prompt engineering pipeline and tested 200+ variations across GPT-4, Claude 3, and Mixtral; documented findings in technical blog post (1,800 views, 60 GitHub stars on accompanying code)
- Constructed fine-tuning datasets from proprietary e-commerce logs; applied LoRA and QLoRA techniques to adapt models for merchandising domain
- Presented "Agents in Retail: Bridging Search and Language" at Seattle AI Meetup (August 2024, 120+ attendees)

Data Science Intern, Clearwater Retail Solutions

Bend, OR | June 2022 – June 2023

- Developed embedding-based product clustering system for category generation; reduced downstream labeling effort by 35% on test sets
- Assisted with LLM API integrations for content generation workflows

SKILLS

LLMs (LLaMA, GPT-4, Claude, Mixtral) | RAG systems | Multi-agent frameworks | Fine-tuning (LoRA, QLoRA) | Prompt engineering | Vector databases (Weaviate, Pinecone) | LangGraph | Python | Transformers | Evaluation methodologies